

UART BLUETOOTH CAR CONTROLLER

RTL-to-GDSII Implementation

A Digital VLSI Physical Design Project

Project	UART-BlueTooth-Car-Controller-RTL-to-GDSII
Design type	Sequential / clocked RTL design
Technology	Sky130
Implementation flow	OpenLane v1.0.2
Top module	uart_bt_car_top
OpenLane run	RUN_2026.10.09_09.16.45
Main function	UART command decoding for motor-control outputs
Final deliverable	GDSII layout and physical-design reports

Final Project Report

Prepared using the Bluetooth car controller source report bundle and recorded STA results supplied for this project.

1. Abstract

This project implements a UART-controlled Bluetooth car controller as a sequential digital design and carries it through an RTL-to-GDSII physical-design flow using Sky130 and OpenLane. The digital design receives serial data from a Bluetooth interface through a UART receiver, decodes command bytes, and generates control signals intended for an external motor driver.

The supplied OpenLane run contains extracted static-timing reports for minimum, maximum and nominal RC corners, power reports, check reports, and physical-design artifacts. This report records the actual timing and power values present in the uploaded source PDF; values not shown in that source are explicitly marked as not reported rather than estimated.

2. Project Objectives

- Implement a UART receiver and command decoder for car movement control.
- Support forward, backward, left, right and stop commands.
- Verify RTL behaviour with a testbench and waveform inspection.
- Run the design through synthesis and physical implementation in OpenLane.
- Review timing at minimum, maximum and nominal RC corners.
- Preserve the GDSII, layout data and signoff/STA evidence for project documentation.

3. Design Overview

The top-level module is **uart_bt_car_top**. The design is clocked and includes reset, UART serial reception, command decoding and motor control outputs. The Bluetooth module is an external serial-data source; the RTL handles the digital command-processing portion.

3.1 Main Design Characteristics

Item	Project implementation
Top-level module	uart_bt_car_top
UART receiver	rtl/uart_rx.v
Command decoder	rtl/bt_car_controller.v
Top-level integration	rtl/uart_bt_car_top.v
Clock/reset	Clocked design with reset input
Motor interface	Digital motor-control outputs for external driver
Physical platform	Sky130 standard-cell flow

3.2 Command Mapping

Received command	Intended action
F	Forward
B	Backward
L	Left
R	Right
S	Stop

The command mapping describes the RTL interface. It does not by itself establish that a physical Bluetooth module, motor driver or car was hardware-tested.

4. RTL Verification and Simulation

The project includes a dedicated testbench and a waveform dump for simulation inspection. GTKWave was used to inspect the serial receiver/command-control behaviour, and a KLayout view was retained as physical-layout evidence.

Verification evidence

Evidence	Project file / artifact
UART receiver RTL	rtl/uart_rx.v
Command controller RTL	rtl/bt_car_controller.v
Top-level RTL	rtl/uart_bt_car_top.v
Testbench	simulation/tb_uart_bt_car.v
Waveform dump	simulation/uart_bt_car.vcd
GTKWave screenshot	screenshots/bluetooth_car_gtkwave.png
KLayout screenshot	screenshots/bluetooth_car_klayouts.png

Simulation checks the digital logic before physical implementation. Physical hardware operation should only be claimed if a real Bluetooth module, motor driver and motors have been tested separately.

5. Physical Design Implementation

The recorded OpenLane run is **RUN_2026.10.09_09.16.45** under the design directory **designs/uart_bt_car/**.

5.1 Flow stages

Stage	Purpose
RTL / synthesis	Translate the Verilog hierarchy into standard-cell logic.
Floorplan	Define the core/die region and I/O planning.
Placement	Place the synthesized standard cells.
Clock-tree stage	Clock-related implementation/analysis for this sequential design.
Routing	Create interconnect between placed cells and I/O pins.
Extraction and STA	Evaluate timing with extracted parasitics across RC corners.
Signoff and GDSII	Produce physical verification reports and final layout deliverables.

5.2 Recorded artifacts

The uploaded report bundle references final signoff reports, extracted RC timing reports, routing-stage reports, metrics, and a final GDSII file for `uart_bt_car_top`. The exact standard-cell count and total cell area are not stated in the supplied STA PDF and are therefore not inserted here.

6. Static Timing Analysis (STA)

The supplied STA PDF identifies three extracted RC timing summaries: rcx_min_sta.summary.rpt, rcx_max_sta.summary.rpt and rcx_nom_sta.summary.rpt. The recorded TNS and WNS values are 0.00 in all three summaries.

6.1 Multi-corner timing summary

Report / corner	TNS	WNS	Worst setup slack
rcx_min_sta / Fastest	0.00	0.00	2.93
rcx_max_sta / Slowest	0.00	0.00	2.87
rcx_nom_sta / Typical	0.00	0.00	2.90

6.2 Interpretation

- TNS = 0.00 and WNS = 0.00 are explicitly recorded in each of the three summary reports.
- The worst setup slack is positive in each summary: 2.93, 2.87 and 2.90 respectively.
- The uploaded summary files do not provide a worst-hold-slack field in the extracted summary text; a hold-slack value is therefore not asserted in this report.
- The detailed check reports contain path-by-path timing data for reset recovery, register-to-register paths and output timing.

Slack values are transcribed as shown in the source report; the OpenLane timing units are conventionally nanoseconds.

7. Power Analysis and Timing Checks

The power report files provide separate Fastest, Slowest and Typical corner estimates. The total power values below are transcribed from the Total rows of the respective reports.

7.1 Total power by RC report

STA report	Fastest total (W)	Slowest total (W)	Typical total (W)
rcx_min_sta.power.rpt	7.98e-04	5.44e-04	6.88e-04
rcx_max_sta.power.rpt	8.12e-04	5.54e-04	7.00e-04
rcx_nom_sta.power.rpt	8.06e-04	5.50e-04	6.95e-04

7.2 Check-report findings

Check	Recorded source result
Negative-slack query (slack_max -0.01)	No paths found
Maximum slew violations	0
Maximum capacitance violations	0
Maximum fanout violations	1
Fanout pin	clkbuf_2_2__f_clk/X
Fanout limit / observed	10 / 14
Fanout slack	-4 (violated)
Unannotated drivers	0
Partially unannotated drivers	0

8. Physical Signoff and Final Deliverables

The uploaded STA bundle includes reports associated with signoff, routing and physical-design verification. It contains a DRC database/report, XOR report, antenna-violation report, final implementation report, and extracted timing reports. The supplied PDF does not expose a complete verified summary of final DRC/LVS/XOR counts, so this report does not claim zero violations or zero errors without that summary.

8.1 Report and artifact inventory visible in the uploaded bundle

Artifact / report	Purpose
uart_bt_car_top.gds	Final layout database
uart_bt_car_top.lef	Abstract layout exchange format
rcx_min_sta.*	Minimum-corner extracted timing/power/check reports
rcx_max_sta.*	Maximum-corner extracted timing/power/check reports
rcx_nom_sta.*	Nominal-corner extracted timing/power/check reports
35-xor.rpt	Layout comparison report
drc.rdb / drc.drc	DRC-related output
41-antenna_violators.rpt	Antenna-check output
uart_bt_car_top.rpt	Final implementation report
metrics.csv	Flow metrics

Use the original signoff summary reports when stating final pass/fail counts in a submission or presentation.

9. Project Repository Organization

The Bluetooth car controller is maintained as a separate project from the UART transmitter project. The STA values in this report are from the Bluetooth car controller run and are not copied from the UART transmitter report.

Directory / file	Purpose
rtl/	UART receiver, command controller and top-level Verilog
simulation/	Testbench and waveform dump
screenshots/	GTKWave and KLayout evidence
openlane/uart_bt_car/	OpenLane design configuration and source
results/final/	Final GDSII/LEF and final deliverables
results/sta/	Timing reports organized by corner
results/signoff/	DRC/LVS/XOR/antenna and signoff report files
docs/	Final report and numerical results summary
README.md	Project description and usage instructions

Keep the Bluetooth car report, STA reports, screenshots and GDSII artifacts in this repository. Do not place UART-transmitter STA numbers into this project.

10. Technical Outcomes and Conclusion

The Bluetooth car controller project demonstrates a clocked RTL design that receives serial command data, decodes motion commands, and generates digital motor-control outputs. The recorded OpenLane run provides minimum, maximum and nominal RC-corner STA summaries and power estimates.

Recorded timing highlights

Metric	Bluetooth car controller result
TNS	0.00 in min/max/nom summary reports
WNS	0.00 in min/max/nom summary reports
Worst setup slack	2.93 (min), 2.87 (max), 2.90 (nom)
Fanout violations	1 reported; fanout 14 against limit 10
Unannotated drivers	0
Power	Corner-specific values in Section 7

Conclusion

The project has been taken through an RTL-to-GDSII implementation flow and has extracted multi-corner timing/power reports available for review. The STA results show zero TNS/WNS in the three supplied summaries and positive worst setup slack values. The checks report also identifies one maximum-fanout violation, which should be documented rather than hidden. The final report is based on the uploaded Bluetooth car controller evidence; missing area, cell-count, hold-slack and complete signoff counts have not been fabricated.

End of Report